BRSM Project Mid-Report

Experiment: Sentence Memorability

Team: BRSM Team_NNK

Members: Ritvik (2022111034) | Revanth (2022101049) | Keshava (2022101123)

1. Introduction

Understanding why some sentences are remembered more easily than others is an important question in the study of human memory and language processing. In everyday communication, people encounter information in the form of sentences rather than isolated words. The memorability of a sentence may depend on the types of words it contains as well as how those words are arranged within the sentence. Words that are vivid, concrete, or emotionally engaging may make a sentence easier to remember, while more abstract words may reduce memorability. Additionally, the way a sentence is framed or structured could influence how it is processed and stored in memory. Exploring sentence memorability can therefore help reveal how lexical content and sentence structure interact in shaping human memory.

2. Experiment Design

The study used a continuous recognition paradigm with a dual-response design to investigate sentence memorability. Sentences were presented one at a time, and participants indicated whether a sentence had appeared before. If they believed a sentence was repeated, they pressed the spacebar to provide an Information Recognition (IR) response. Immediately afterward, a Yes/No prompt asked them to confirm their decision through a Word Recognition (WR) judgment. Reaction times (RT) for both IR and WR responses were recorded.

Sentences followed a consistent Subject–Verb–Object (S–V–O) structure. Verbs with medium concreteness were selected so that memorability manipulations primarily affected the subject and object nouns. Word memorability was manipulated across four types: HH (high–high), HL (high–low), LH (low–high), and LL (low–low). Each sentence was presented in either Active or Passive voice, resulting in eight experimental conditions.

Participants first completed a practice phase, followed by three experimental blocks containing experimental sentences, filler items, and validation sentences used as attention checks. Rest breaks were provided between blocks. This design enabled the collection of IR responses, WR accuracy, and reaction times.

3. Hypotheses

- **H1 (Voice Effect on IR):** Sentence voice will influence sentence memorability in the

Information Recognition task, such that sentences presented in active voice will produce higher recognition performance than those in passive voice.

- **H2 (Voice Effect on WR):** Sentence voice will influence Word Recognition accuracy, with active voice sentences expected to yield higher WR accuracy than passive voice sentences due to easier processing and encoding.
- **H3 (Word Memorability Effect on IR):** Sentence memorability in the Information Recognition task will vary according to the memorability of the constituent nouns. Sentences containing high-memorability words (HH) are expected to produce higher recognition performance than mixed (HL, LH) or low-memorability (LL) sentences.
- **H4 (Word Memorability Effect on WR):** Word Recognition accuracy will differ across word memorability conditions, with sentences containing high-memorability nouns leading to higher WR accuracy compared to sentences containing low-memorability nouns.

4. Dataset Description

The data we have collected consists of logfiles generated while participants performed the experiments. Each log file contains the following information:

- **Event:** Type of events that occurred during the experiment (e.g., Sentence shown, IR pressed, WR pressed, Missed, Validation IR pressed, Rest Phase started).
- **Stimulus:** The sentence presented to the participant, represented by a sentence ID (e.g., HH_112_A).
- **isValidation:** Indicates whether the presented sentence is a validation sentence (True/False).
- **isRepeat:** Indicates whether the sentence is a repetition of a previously shown sentence (True/False).
- **Button:** The button pressed by the participant in response to the stimulus.
- **Accuracy_IR:** Indicates whether the participant correctly identified if the sentence was repeated in the IR task.
- **Accuracy_WR:** Indicates whether the participant correctly identified whether the sentence voice was active or passive in the WR task.
- **Reaction_time_IR:** Time taken to respond in the IR task, measured from stimulus presentation to response.
- **Reaction_time_WR:** Time taken to respond in the WR task, measured from stimulus presentation to response.

5. Data Preprocessing

As mentioned earlier, the dataset contains the logfiles for each participant with the information explained in section 4.

A. Blocks Formation and Attribute Manipulation

We divided the logfile of each participant into 3 files representing each block of sentences as designed in the experiment, along with some manipulations on the data:

- In the logfiles, events were recorded sequentially. So, information about a single sentence (whether IR or WR was pressed) was stored in different tuples. We modified this by removing the Event column and adding IR_Event and WR_Event for each sentence in that block.
- The Timestamp column was removed, a block attribute was added representing which block the sentence belongs to, and CR_IR, CR_WR attributes were removed as they had only one unique value, adding no information.

B. Removing Invalid Datapoints

Each block included validation sentences to identify inattentive participants. In the dataset, these events are marked in the isValidation column. A block was considered valid only if the participant met a predefined attention criterion:

- Correct Validation IRs > (Wrong Validation IRs ÷ 2) + Missed Validation IRs
- *(Where Correct = accurately identified, Wrong = incorrectly identified, Missed = no response)*

Among all the blocks from the dataset across all participants, we identified 6 sentence blocks as invalid. These blocks were removed from the dataset.

C. Transforming the Data

Our data structure is now composed of different block files with each tuple having aggregated information about each sentence (IR/WR accuracy, reaction times, etc.). Since we are studying the sentences, we aggregated all block files into a single master_dataset.csv. This was further reduced into stimulus_summary.csv, which aggregates the information per unique sentence. Attributes include: Stimulus, sentence_type, voice, Num_participants, Num_Correct_IR_Hit, Num_correct_IR_Miss, Num_wrong_IR, Num_Missed_IR, Num_Correct_WR, Num_Wrong_WR, Num_WR. These are used to calculate the variables mentioned in the next section.

D. Variables Formulation

Using the stimulus_summary.csv, we formulated the following derived metrics to test our hypotheses:

1. **Corrected Memorability Score (CMS):** This metric acts as an adjusted measure of recognition performance, accounting for response biases by calculating the difference between the hit rate and the false alarm rate. In terms of True Positives (**TP**), True Negatives (**TN**), False Positives (**FP**), and False Negatives (**FN**), the formula is:

$$CMS = \text{Hit Rate} - \text{False Alarm Rate} = \frac{TP}{TP + FN} - \frac{FP}{FP + TN}$$

   We calculate this for both the IR task (CM_IR) and the WR task (CM_WR).
2. **Accuracy:** A straightforward calculation of the proportion of correct responses out of the total valid trials. The formula is:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

This is determined separately for Information Recognition (IR_Accuracy) and Word Recognition (WR_Accuracy).

Because we apply both CMS and Accuracy to both IR and WR tasks, we have a total of 8 primary metrics for our analysis.

6. Descriptive Statistics

Note on Data Distribution: Each participant was shown 18 sentences per type and 36 sentences per voice. Overall, each category contains 49 unique sentences per sentence type and 98 per voice.

6.1. Overall Statistics

Metric	IR_Accuracy	WR_Accuracy	CM_IR	CM_WR
mean	0.895	0.744	0.791	0.247
std	0.067	0.143	0.133	0.478
min	0.607	0.167	0.271	-0.667
25%	0.857	0.655	0.714	-0.200
50%	0.905	0.750	0.810	0.167
75%	0.941	0.843	0.882	0.714
max	1.000	1.000	1.000	1.000
var	0.004	0.020	0.018	0.228

6.2. Statistics by Sentence Type

A. IR_Accuracy

Sentence Type	mean	std	min	25%	50%	75%	max

HH	0.901	0.059	0.700	0.870	0.904	0.939	1.000
HL	0.896	0.068	0.667	0.857	0.911	0.944	1.000
LH	0.904	0.061	0.607	0.870	0.910	0.944	1.000
LL	0.879	0.075	0.679	0.833	0.885	0.932	1.000

B. WR_Accuracy

Sentence Type	**mean**	**std**	**min**	**25%**	**50%**	**75%**	**max**
HH	0.744	0.131	0.444	0.667	0.750	0.833	1.000
HL	0.747	0.144	0.385	0.650	0.727	0.846	1.000
LH	0.771	0.145	0.250	0.694	0.760	0.871	1.000
LL	0.713	0.146	0.167	0.625	0.714	0.809	1.000

C. CM_IR

Sentence Type	**mean**	**std**	**min**	**25%**	**50%**	**75%**	**max**
HH	0.804	0.121	0.400	0.739	0.808	0.876	1.000
HL	0.797	0.133	0.291	0.718	0.826	0.875	1.000
LH	0.808	0.120	0.271	0.750	0.816	0.889	1.000
LL	0.757	0.152	0.333	0.667	0.781	0.873	1.000

D. CM_WR

Sentence Type	**mean**	**std**	**min**	**25%**	**50%**	**75%**	**max**

HH	0.264	0.472	-0.529	-0.186	0.268	0.694	1.000
HL	0.287	0.472	-0.500	-0.156	0.314	0.683	1.000
LH	0.249	0.542	-0.667	-0.224	0.182	0.755	1.000
LL	0.177	0.436	-0.400	-0.188	0.052	0.636	0.923

6.3. Statistics by Voice

A. IR_Accuracy

Voice	mean	std	min	25%	50%	75%	max
A	0.894	0.068	0.607	0.846	0.905	0.943	1.000
P	0.896	0.066	0.679	0.862	0.905	0.938	1.000

B. WR_Accuracy

Voice	count	mean	std	min	25%	50%	75%	max
A	196.0	0.753	0.138	0.333	0.667	0.750	0.857	1.000
P	196.0	0.735	0.146	0.167	0.642	0.735	0.833	1.000

C. CM_IR

Voice	count	mean	std	min	25%	50%	75%	max
A	196.0	0.789	0.133	0.271	0.700	0.800	0.882	1.000
P	196.0	0.794	0.133	0.333	0.732	0.815	0.877	1.000

D. CM_WR

Voice	count	mean	std	min	25%	50%	75%	max
A	65.0	0.215	0.464	-0.600	-0.200	0.167	0.636	1.000

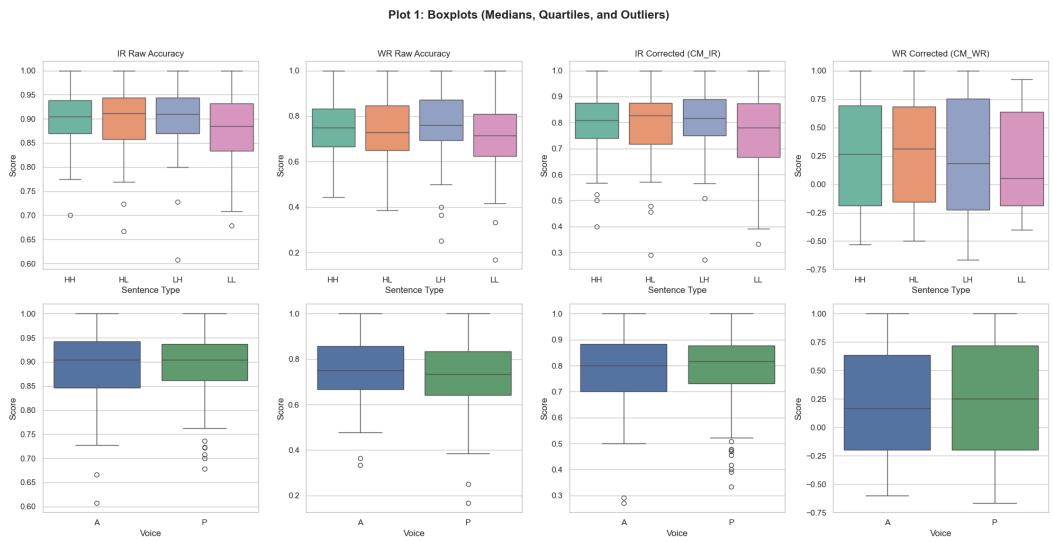
P	75.0	0.274	0.490	-0.667	-0.200	0.250	0.721	1.000

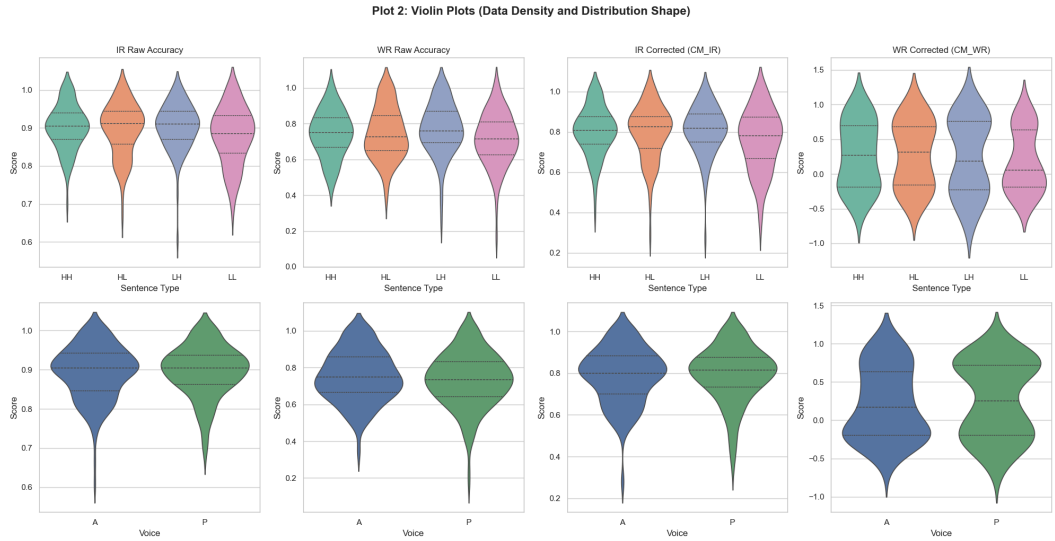
6.4. Plots Visualization

To better understand the distributions of our data, we generated two primary visualizations for the metrics mentioned above:

Box Plots: These were utilized to identify the interquartile ranges, medians, and to detect any prominent outliers across different sentence voices and types.

Violin Plots: These helped capture the probability densities of the continuous variables, highlighting variations in data clustering, particularly revealing the multi-modal nature of some metric distributions.



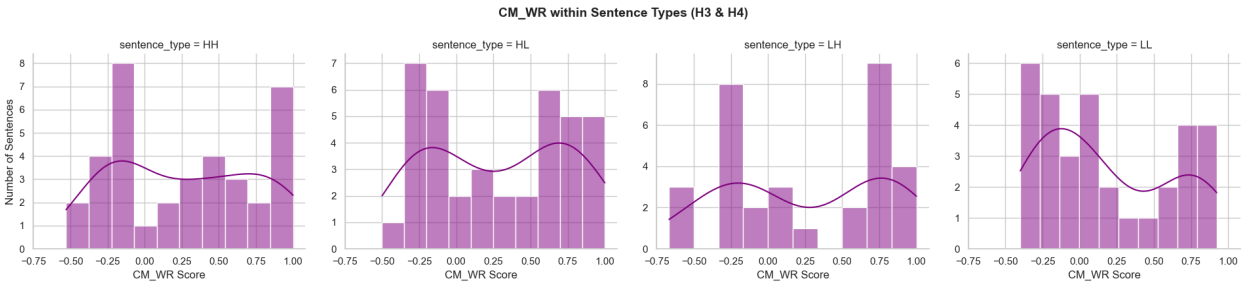


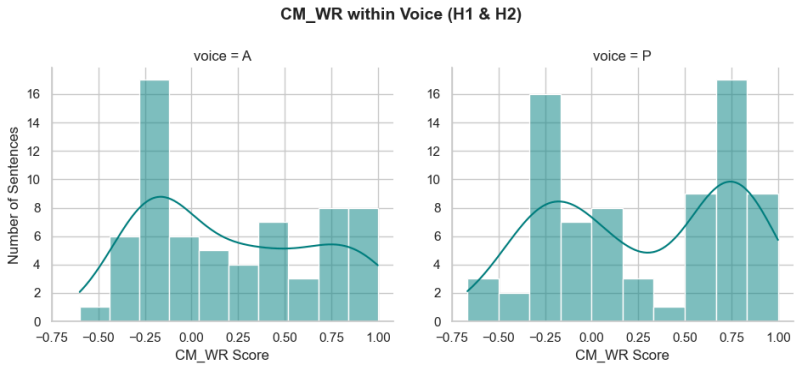
7. Exploratory Analysis

Before moving to inferential statistics, we conducted normality checks to determine the appropriate statistical methods for evaluating our hypotheses.

Upon analyzing the distributions, we observed that **IR Accuracy** and **WR Accuracy** conform approximately to a normal distribution. Because these assumptions are met, we can comfortably rely on parametric tests (specifically, Generalized Linear Mixed Models or GLMMs) to investigate the fixed effects of voice and memorability on accuracy.

However, the Corrected Memorability Scores behave differently. The **CM_WR (Corrected Memorability for Word Recognition)** metric is strictly non-normal. Similarly, **CM_IR (Corrected Memorability for Information Recognition)** displays bimodal tendencies across certain sentence types, suggesting sub-populations within participant responses. Due to the violation of normality in these variables, we cannot use standard parametric modeling. Instead, we will deploy non-parametric tests, such as the Kruskal-Wallis H-test, to detect statistically significant differences among the CM_WR and CM_IR distributions.





8. Conclusion and Future Work

Based on our preliminary descriptive and exploratory analysis, we observe several trends that provide initial evaluations of our hypotheses:

- **Voice Effects (H1 & H2):** Contrary to our expectations in H1, voice has a negligible effect on Information Recognition (IR) accuracy, with Passive voice (0.896) performing practically identically to Active voice (0.894). However, supporting H2, Word Recognition (WR)

accuracy is slightly higher for Active sentences (0.753) than Passive sentences (0.735), hinting that active voice facilitates easier deep encoding.

- **Word Memorability Effects (H3 & H4):** The descriptive statistics offer robust preliminary support for H3 and H4. Sentences composed entirely of low-memorability words (LL) consistently yielded the worst performance across both Information Recognition (IR Accuracy = 0.879, CM_IR = 0.757) and Word Recognition (WR Accuracy = 0.713, CM_WR = 0.177). Meanwhile, sentences containing high-memorability words (HH, HL, LH) resulted in notably higher recognition scores. Interestingly, the LH sentence type showed the highest WR Accuracy (0.771).

Moving forward, we will formalize these observations utilizing GLMMs and Kruskal-Wallis tests to rigorously ascertain if these differences are statistically significant.

9. References

1. Snodgrass, J. G., & Corwin, J. (1988). Pragmatics of measuring recognition memory. Journal of Experimental Psychology: General.
2. Swets, J. A., Tanner, W. P., & Birdsall, T. G. (1961). Decision processes in perception. Psychological Review.
3. Begg, I. (1971). Recognition memory for sentence meaning and wording. *Journal of Verbal Learning and Verbal Behavior*, 10(2), 176-181.
4. Bregman, A. S., & Strasberg, R. (1968). Memory for the syntactic form of sentences. *Journal of Verbal Learning and Verbal Behavior*, *7*(2), 396–403.

10. Repository

All code for data preprocessing, exploratory analysis, and visualization can be found at our project repository: Github Link *(Data is not included since it is a public repo).*

11. Contributions

- **Ritvik -** Stimulus level data and metrics, Report writing and Dataset Understanding

- **Revanth -** Descriptive statistics and normality checks, Report writing and Dataset Understanding

- **Keshav -** Loading and compiling to master dataset, Report writing and Dataset Understanding